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AI CAREERS FOR WOMEN

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Hospital Patient Records & Health Metrics Dashboard

Project Domain

[Power BI with Copilot]

A Collaborative Initiative

Equipping students with practical skills in healthcare data analytics.

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3. Problem Statement

Hospitals generate a large volume of patient data every day through various sources such as patient registrations, medical records, laboratory reports, billing systems, and treatment histories. However, this data is often unorganized, incomplete, duplicated, or inconsistent due to manual data entry, different data collection systems, and human errors. Because of these issues, hospitals face difficulty in accurately analyzing patient information and monitoring important health metrics.

Healthcare administrators and hospital management rely on accurate and well-structured data to make critical decisions related to patient care, hospital efficiency, resource allocation, and treatment outcomes. When the data is not properly cleaned or structured, it becomes difficult to generate reliable insights, which can affect hospital performance and decision-making.

Another challenge in the healthcare domain is the lack of effective data visualization and reporting systems. Raw hospital data stored in spreadsheets or databases is difficult to interpret without proper dashboards or analytics tools. This creates a gap where valuable healthcare insights remain unused.

This project focuses on solving these challenges by performing patient records data cleaning and developing a Health Metrics Dashboard. Using tools such as Microsoft Excel and Power BI, the project will organize and clean hospital patient records, remove inconsistencies, and transform the data into meaningful insights through visual dashboards.

The solution will help hospital administrators, healthcare analysts, and medical management teams better understand patient trends, treatment statistics, hospital performance, and key healthcare metrics. By converting raw hospital data into structured and visualized information, the project aims to support better decision-making, improved healthcare management, and enhanced operational efficiency in hospitals.

4. Objectives

The main objectives of this project are to analyze hospital patient data, improve data quality, and create meaningful healthcare insights through a dashboard. The specific objectives of the project are:

• **To collect and understand the hospital patient records dataset** by examining its structure, fields such as patient ID, age, gender, diagnosis, admission details, and identifying data quality issues.

• **To clean and preprocess the hospital data** by removing duplicate patient records, handling missing values, correcting incorrect formats, and standardizing fields to ensure accuracy and reliability.

• **To analyze important healthcare metrics** such as total number of patients, admission rates, discharge statistics, disease distribution, and patient demographics using tools like **Excel formulas, pivot tables, and Power BI measures.**

• **To create a Health Metrics Dashboard** that visually presents hospital performance indicators such as patient trends, department-wise cases, treatment statistics, and other key healthcare insights.

• **To integrate AI tools such as Microsoft Copilot or AI Chatbots** to assist in data analysis, report generation, and explanation of healthcare insights, improving efficiency and productivity.

• **To transform raw hospital patient data into meaningful visual insights** that can help hospital administrators and healthcare managers make better data-driven decisions.

• **To document the complete project workflow**, including data cleaning steps, analysis process, dashboard creation, and insights, and prepare the final report and submission materials.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	Data Collection & Exploration	The hospital patient records dataset will be collected in Excel/CSV format from a reliable source such as Kaggle or an academic dataset. The dataset will be explored to understand the number of records, field names (Patient ID, Age, Gender, Disease, Admission Date, Discharge Date, Department, etc.), and data types. Initial exploration will help identify data quality issues such as missing values, duplicate entries, or incorrect formats.
2	Data Cleaning & Preprocessing	The dataset will be cleaned using Microsoft Excel or Power Query. This step includes removing duplicate patient records, handling missing values, correcting incorrect data formats (dates, numbers, text), and standardizing fields such as gender, department names, or diagnosis categories. These steps will ensure the dataset becomes accurate and suitable for analysis..

3	Analysis & AI Integration	After cleaning the dataset, analysis will be performed using Excel formulas, Pivot Tables, or Power BI measures to identify key healthcare insights. Important metrics such as total number of patients, department-wise admissions, disease distribution, patient demographics, and discharge statistics will be calculated. AI tools such as Microsoft Copilot or AI Chatbots may also be used to generate summaries, explanations, or insights from the analyzed data.
4	Visualization / Output	A Health Metrics Dashboard will be created using Power BI or Excel charts. The dashboard will include visualizations such as bar charts, pie charts, KPI cards, and slicers to display important healthcare indicators like patient trends, disease distribution, and department performance. These visualizations will help hospital administrators easily understand key insights from the data.
5	Documentation & Submission	Finally, the entire project process will be documented in a structured report including problem statement, objectives, methodology, data cleaning steps, analysis results, and dashboard explanation. All project files, datasets, and reports will be uploaded to GitHub, and a video demonstration explaining the dashboard and analysis will be recorded and submitted for evaluation.

6. Resources and Timeline

A. Hardware & Software Requirements

The following hardware and software tools will be used to complete the project successfully:

• Laptop or PC — Minimum Intel Core i3 processor, 4 GB RAM (8 GB recommended) for smooth data processing and dashboard creation.

• Operating System — Windows 10 / Windows 11 or macOS to support Excel and Power BI tools.

• Microsoft Excel (Microsoft 365 / Excel 2019 or later) — Used for data cleaning, preprocessing, formulas, and pivot table analysis of hospital patient records.

• Power BI Desktop (Latest Version) — Used to build the Health Metrics Dashboard with interactive charts, KPI cards, and visualizations.

• Microsoft Copilot — Used to assist in data analysis, generating insights, and creating explanations for healthcare metrics where applicable.

• AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — Used to help with content generation, summaries, and explanation of healthcare insights during the project documentation.

• Internet Connection — Required for accessing AI tools, dataset sources, and online resources.

• Web Browser (Google Chrome or Microsoft Edge – Latest Version) — Used for accessing GitHub, dataset platforms, and AI tools.

B. Dataset

The dataset used in this project is Hospital Patient Records Dataset.

• The dataset contains information related to patient details, admission records, hospital departments, diagnosis, treatment information, and discharge status.

• It includes fields such as Patient ID, Age, Gender, Disease/Diagnosis, Admission Date, Discharge Date, Department, and Treatment Details.

• This dataset will be used to perform data cleaning, analysis, and visualization of healthcare metrics.

• The dataset may be obtained from Kaggle, Edunet resources, or a publicly available healthcare dataset, and it will be stored in Excel format for analysis and dashboard creation.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding of hospital patient records such as patient ID, age, gender, diagnosis, admission and discharge details.	*Week 1*
2	Data cleaning and preprocessing including removing duplicate patient records, handling missing values, correcting data formats, and validating the dataset for analysis.	*Week 2*
3	Data analysis and AI tool integration using Excel formulas, Pivot Tables, Power BI measures, and AI tools to generate healthcare insights such as	*Week 2 – 3*

	patient trends, department-wise cases, and disease distribution.	
4	Development of the Health Metrics Dashboard with charts, KPI cards, filters, and visualizations to present hospital performance indicators clearly.	*Week 3*
5	Final report writing, GitHub repository upload, dashboard documentation, and recording of the project demonstration video for submission.	*Week 4*

7. Expected Outcomes / Conclusion

This project aims to transform raw hospital patient data into structured, meaningful insights that can support better healthcare management and decision-making. By the end of the project, the following outcomes are expected:

• A cleaned and validated hospital patient dataset where all preprocessing steps such as removing duplicates, handling missing values, correcting formats, and standardizing fields are clearly documented. A comparison of the dataset before and after data cleaning will also be included.

• A Health Metrics Dashboard created using Excel or Power BI that visually presents important healthcare indicators such as total number of patients, department-wise admissions, disease distribution, patient demographics, and discharge statistics.

• Clear analytical insights derived from hospital data that help understand patient trends, hospital workload, and key performance indicators related to healthcare services.

• Integration of AI tools such as Microsoft Copilot or AI Chatbots to assist in generating insights, explanations, summaries, or documentation related to the hospital data analysis. The prompts used and their outputs will be properly documented.

• A complete project repository on GitHub containing the cleaned dataset, dashboard files, project report, and all supporting materials used during the project.

• A final project report and demonstration video explaining the data cleaning process, dashboard features, and insights generated from the hospital patient records dataset.

• Development of Health Metrics Visualizations including charts and KPI indicators that show patient admissions, disease distribution, department-wise cases, and patient demographics, helping hospital management easily monitor hospital performance and patient trends through an interactive Power BI Health Metrics Dashboard.

8. References

List all references, tutorials, and documentation used in the project.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
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